Interpreting Odd Ratios in Logistic Regression

OmaymaS

Introduction

Interpreting the logistic regression's coefficients is somehow tricky. Looking at some examples beside doing the math helps getting the concept of odds, odds ratios and consequently getting more familiar with the meaning of the regression coefficients. The following examples are mainly taken from IDRE UCLE FAQ Page (http://www.ats.ucla.edu/stat/mult_pkg/faq/general/odds_ratio.htm) and they are recreated with R.

Probability, Odds and Log of Odds

Let's say that the probability of success is $p = 0.8$, then the probability of failure is $1 - p = 0.2$. The odds of success is $\frac{p}{1-p} = \frac{0.8}{1-0.8} = 4$, i.e. the odds of success is 4 to 1 and the odds of failure is 0.25 to 1.

Note that:

- **Probability** ranges from 0 to 1
- **Odds** range from 0 to ∞
- **Log Odds** range from $-\infty$ to ∞

That is why the log odds are used to avoid modeling a variable with a restricted range such as probability.

Logistic Regression

Logistic regression models the the logit-transformed probability as a linear relationship with the predictor variables as follows:

$$logit(p) = log(\frac{p}{1-p}) = \beta_0 + \beta_1 x_1 + \ldots + \beta_x x_x$$

or in terms of probabilities:

$$p = \frac{exp(\beta_0 + \beta_1 x_1 + \ldots + \beta_x x_x)}{1 + exp(\beta_0 + \beta_1 x_1 + \ldots + \beta_x x_x)}$$

Examples

The following examples use a dataset (http://www.ats.ucla.edu/stat/mult_pkg/faq/general/sample.csv) that contains 200 observations about students. The binary outcome variable we will use is **hon** which indicates if a student is an honor class or not.

Loading Libraries and Data

```
library(dplyr)
library(knitr)
library(pander)
```

```
hd<-read.csv("honordata.csv",sep=",")

head(hd)
```

```
##   female read write math hon femalexmath
## 1      0   57    52   41   0           0
## 2      1   68    59   53   0          53
## 3      0   44    33   54   0           0
## 4      0   63    44   47   0           0
## 5      0   47    52   57   0           0
## 6      0   44    52   51   0           0
```

Logistic Regression with No Predictor Variables

Here we will start with a simple model without any predictors:

$$logit(p) = \beta_0$$

Fitting The Model

```
#fit the model without any predictor
f0<-glm(hon~1,data = hd,family = binomial)

summary(f0)$coeff
```

```
##             Estimate Std. Error   z value     Pr(>|z|)
## (Intercept) -1.12546    0.16441 -6.845446 7.623779e-12
```

As we can see:

- The intercept= **-1.12546** which corresponds to **the log odds of the probability of being in an honor class** p .